

NATURAL LANGUAGE PROCESSING (PRACTICE)

NLP 252 - LAB1: Lab Introduction



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General Introduction

- The lab sessions take place over 15 weeks (2 periods per week).
- Each week usually consists of two types of exercises: **In-class Exercise** (Deadline: 1 day) and **Home-class Exercise** (Deadline: 1–2 weeks).
- The exercises will include two formats: coding exercises and mathematical exercises.
- Each exercise allows up to 7 late submission days. For every late day, 1 point will be deducted from the total score.

The course consists of 12 lab sessions, 1 final lab exam, and 2 additional lecture sessions (the content may vary depending on the semester). The grading policy is defined as follows:

- The lab grade is calculated as the harmonic mean of the final exam score and the average score of the lab exercises.
- If **the lab grade is below 4**, it will be considered as 0, and the student will fail the course.
- If a student **misses 5 or more lab exercises**, the lab grade will be considered as 0, and the student will fail the course.

Final Lab Exam Structure

- The exam is conducted using Safe Exam Browser (SEB). Students are required to install SEB for the exam session.
- Number of questions: 40 multiple-choice questions and 1 written question.
- Duration: 90 minutes.
- Permitted materials: pens, rulers, calculators, and blank scratch paper.
- Students may bring only one handwritten A4 sheet; printed materials are not allowed.

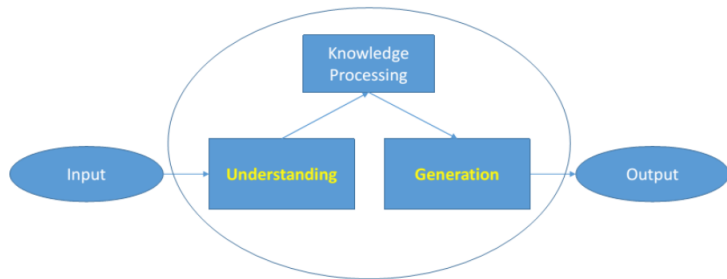
Content Overview

- Preprocessing
- Language Models
- Vector Semantics and Embeddings
- Linear Models
- Optimizers
- Neural Networks
- Recurrent Neural Networks
- Sequence-to-Sequence Models
- Transformer Architectures
- Applications

Natural Language Processing

Introduction

NLP (Natural Language Processing) is the field of study and application aimed at enabling computers to understand and generate natural language.

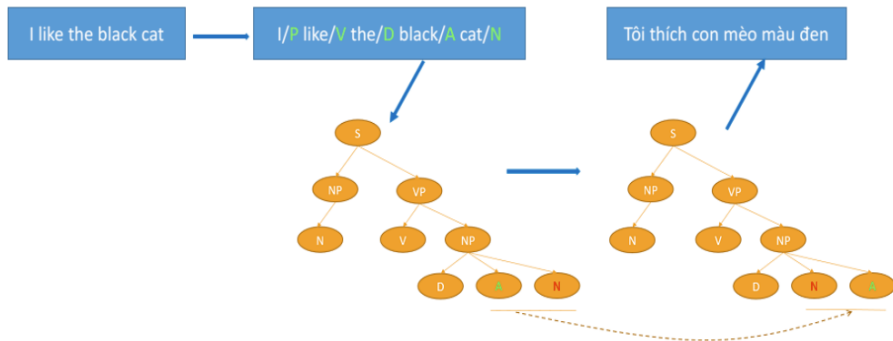


What makes language hard

- 1 Ambiguity
- 2 Context Dependency
- 3 Cultural and Regional Differences
- 4 Background Knowledge
- 5 Massive amounts of unstructured and semi-structured data with new challenges

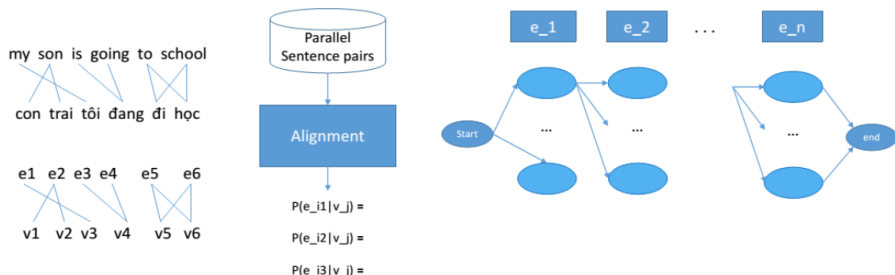
History of NLP

1 From 1950s -> 1980s: Rule-based Approach + experts



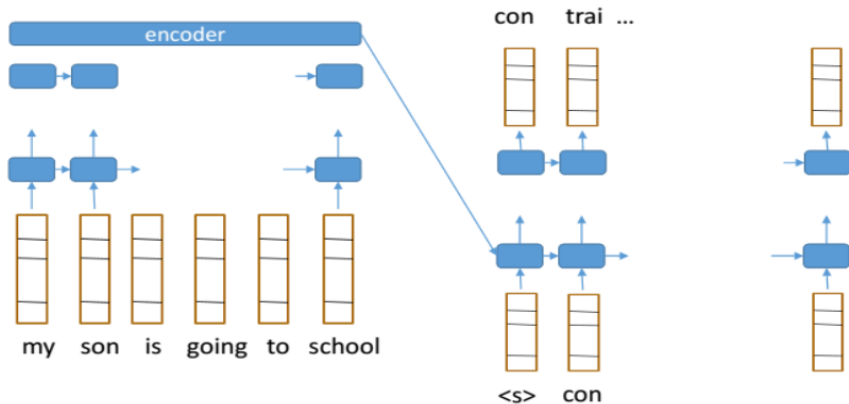
History of NLP

2 Late 1980 -> 2000s: Applying Statistical Machine Learning



History of NLP

3 Late 2010s: Applying Deep Learning: Representation learning



Natural Language Processing Task

- **Text Classification:** Sentiment analysis, Spam detection,
- **Machine Translation:** Translating text from one language to another
- **Text Summarization:** Generating a concise summary of a longer text
- **Question Answering:** Answering questions based on a given context or document

Natural Language Processing Task

- **Language Modeling:** Predicting the next word or sequence in a sentence
- **Text Generation:** Generating coherent and contextually relevant text
- **Topic Modeling:** Discovering the abstract topics that occur in a collection of documents.
- **Word Embedding:** Learning continuous vector representations of words
-

Traditional Methods

- **Rule-Based Systems:**
 - Grammar rules
 - Handcrafted features
- **Statistical Models:**
 - N-grams
 - Hidden Markov Models (HMMs)
 - Conditional Random Fields (CRFs)
- **Bag of Words (BoW):**
 - Simple representation of text
 - Ignores word order
- **TF-IDF (Term Frequency-Inverse Document Frequency):**
 - Weighs the importance of words
 - Widely used in information retrieval

Modern Methods

- **Word Embeddings:**

- Word2Vec, GloVe, FastText
- Capture semantic relationships between words

- **Sequence Models:**

- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory (LSTM)
- Gated Recurrent Units (GRUs)

- **Transformers:**

- BERT (Bidirectional Encoder Representations from Transformers)
- GPT (Generative Pre-trained Transformer)
- T5 (Text-to-Text Transfer Transformer)

- **Pre-trained Language Models:**

- Fine-tuning for specific tasks
- Transfer learning

Important Libraries

- **nltk:** For preprocessing
- **word_tokenize:** For Vietnamese preprocessing
- **Stanford NLP Core:** For parsing
- **Sklearn:** model_selection, metric and train_test_split
- **Word2Vec:** Word Embedding
- **Pytorch, TensorFlow:** Deep learning Framework
- **Transformers, Huggingface:** Pretrained models

Example Math Exercise

Problem

Suppose we have a collection of 3 documents document 1 says "lions eat fat cats"; document 2 says "cats eat fat mice"; document 3 says "mice eat fat cheese". Compute the cosine similarity of document 1 and document 2 with equal TF weighting:

- ☐ 0.25
 ☐ 0.0
 ☐ 0.5
 ☒ 0.75
 ☐ 1.0

Problem

Consider the following corpus of 3 sentences:

- I am here
- Who am I
- I would like to go

Calculate $P(\text{here I am})$ assuming a bi-gram language model.

- ☐ $1/2$
☐ $2/3$
☒ $1/3$
☐ 1

Example Code Exercise

- Get text from web data, pdf, word, txt.
- Check for plagiarism by word embedding and similarity scores.
- Building a spell-checking model based on n-grams
- Labeling parts of speech with a Hidden Markov Model (HMM)
- Training Sequence Model.
- Finetune Pretrained Model.

ANY QUESTIONS?

THANKS FOR
LISTENING!